Cross Product

Section 11.4

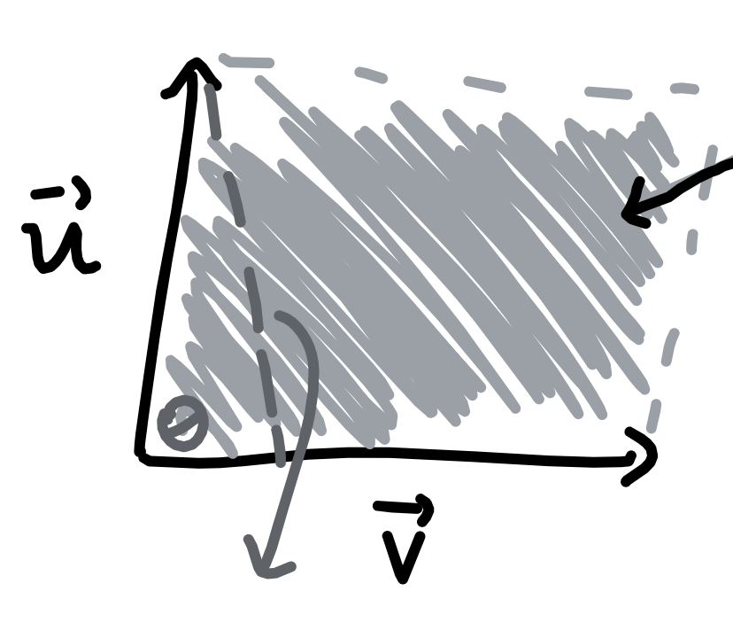
8 - 21 - 26

A) Determinant of a vector (vector product)

$\vec{u}\cdot\vec{v} = \#$ $\quad\vec{u}\times\vec{v}$ = vector

1) $\perp$ to both vectors

2) $\|\vec{u}\times\vec{v}\|$ area of a parallelogram



$= \|v\|\cdot\|u\|\cdot\sin\theta$

3) $\|\vec{u}\times\vec{v}\| = 0 \quad \vec{u}\,/\!/\,\vec{v}$

4) $\vec{a}\times\vec{b} = -(\vec{b}\times\vec{a})$

ex) find a×b

$a = \langle 1, 3, 4\rangle$

$b = \langle 2, 7, -5\rangle$

ⓘ ~~j~~ ~~k~~ $\quad$ ~~i~~ ⓙ ~~k~~ $\quad$ ~~i~~ ~~j~~ ⓚ

$a=\langle 1,3,4\rangle \quad a=\langle 1,3,4\rangle \quad a=\langle 1,3,4\rangle$

$b=\langle 2,7,-5\rangle \quad b=\langle 2,7,-5\rangle \quad b=\langle 2,7,-5\rangle$

$a\times b = (-15-28)i - (-5-8)j + (7-6)k$

$= \boxed{\langle -43, 13, 1\rangle}$

thus, $b\times a = \langle 43, -13, -1\rangle$

ex) Area of Δ w/ vertices

$P(1,4,6)\ Q(-2,5,-1)\ R(1,-1,1)$

$\overrightarrow{PQ} = \langle -3, 1, -7\rangle$

$\overrightarrow{PR} = \langle 0, -5, -5\rangle$

$\overrightarrow{PQ}\times\overrightarrow{PR} = (-5-35)i - (15)j + (15)k$

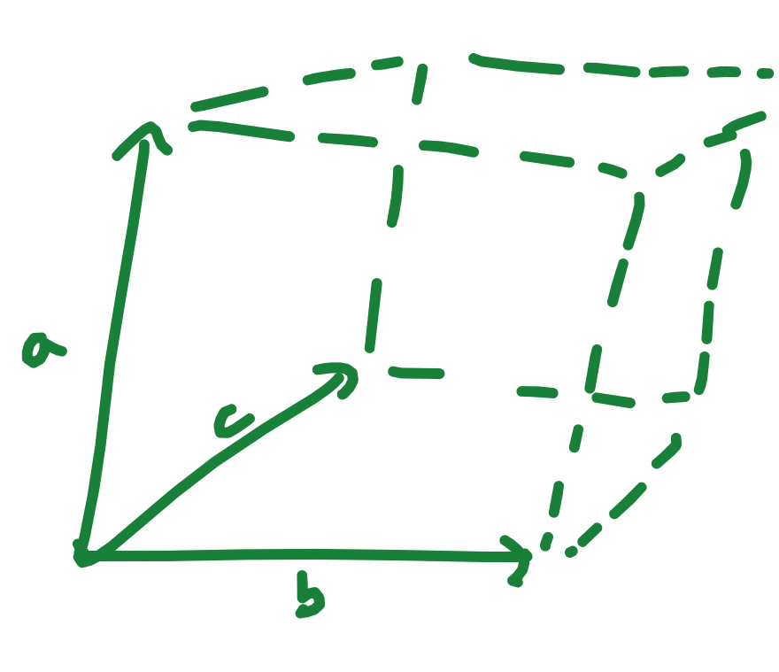
$= \langle -40, -15, 15\rangle = -5\langle -8, 3, 3\rangle$

$\|\overrightarrow{PQ}\times\overrightarrow{PR}\| = 5\sqrt{82}$

Area $= \boxed{\frac{5}{2}\sqrt{82}}$

ex) Volume of a Parallelopiped

(Scalar triple product)



$|\vec{a}\cdot(\vec{b}\times\vec{c})|$

$|\vec{b}\cdot(\vec{a}\times\vec{c})|$

$|\vec{c}\cdot(\vec{a}\times\vec{b})|$

$a = \langle 3, -5, 1\rangle \qquad \vec{b}\times\vec{c} = \langle 4, -6, -6\rangle$

$b = \langle 0, 2, -2\rangle \qquad a = \langle 3, -5, 1\rangle$

$c = \langle 3, 1, 1\rangle \qquad \vec{a}\cdot(\vec{b}\times\vec{c}) = 12+30-6 = \boxed{36}$

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9) $u = \langle 7, 3, 2\rangle$

$v = \langle 1, -1, 5\rangle$

$u\times v = \langle 17, -33, -10\rangle$

$v\times u = \langle -17, 33, 10\rangle$

$v\times v = \langle 0, 0, 0\rangle$

17) $u\times v = \langle -14, -10, 26\rangle$

$\left\langle \frac{-14}{\sqrt{972}}, \frac{-10}{\sqrt{972}}, \frac{26}{\sqrt{972}}\right\rangle$

25) $\overrightarrow{AB} = \langle 1, 2, 3\rangle$

$\overrightarrow{BC} = \langle -1, -2, -3\rangle$

$\overrightarrow{AD} = \langle 5, 4, 0\rangle$

$\overrightarrow{AB}\times\overrightarrow{AD} = \langle -12, 15, -6\rangle = 3\langle -4, 5, -2\rangle$

$3\sqrt{45} = 9\sqrt{3}$

35)

$u = \langle 2, 0, 1\rangle$

$v = \langle 0, 3, 0\rangle \quad = \boxed{6}$

$w = \langle 0, 0, 1\rangle$

3a)

$\vec{a} = \langle 3, 0, 0\rangle \qquad |\vec{c}\cdot(\vec{a}\times\vec{b})|$

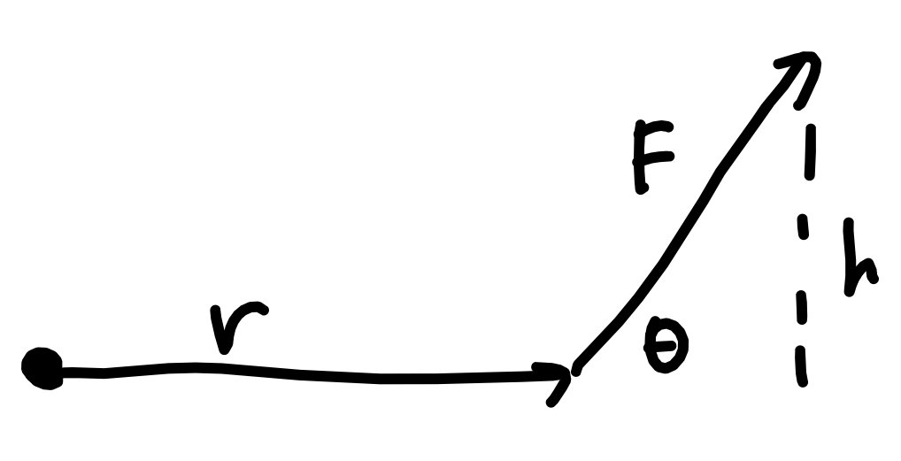
$\vec{b} = \langle 0, 5, 1\rangle \qquad \vec{a}\times\vec{b} = \langle 0, -3, 15\rangle$

$\vec{c} = \langle 2, 0, 5\rangle \qquad \vec{c} = \langle 2, 0, 5\rangle$

$\vec{c}\cdot(\vec{a}\times\vec{b}) = 0+0+75 = \boxed{75}$

B) Torque

τ

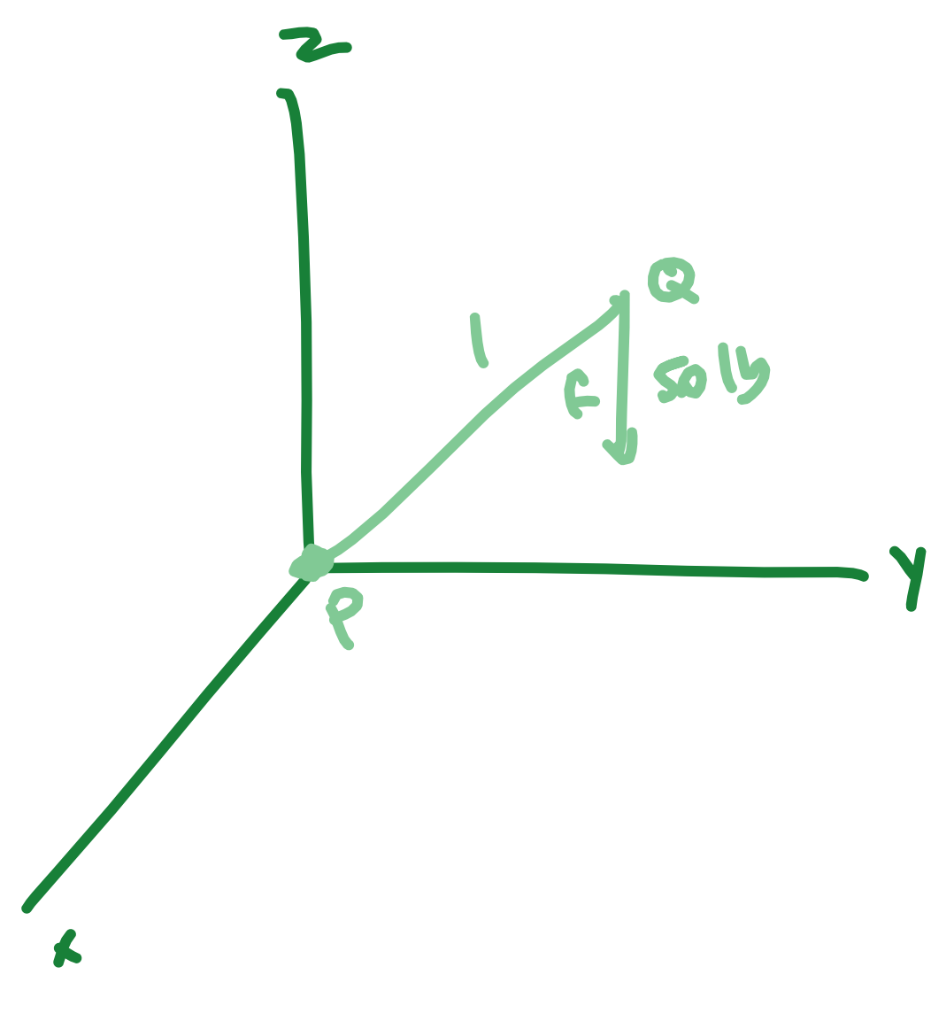


$\|r\|\cdot$ ~~x~~

$\cdot\|F\|\sin\theta$

$\tau = \|r\times F\|$

Ex) A vertical force of 50 lbs is applied to a 1ft lever. Find τ at P if $\theta = 60°$



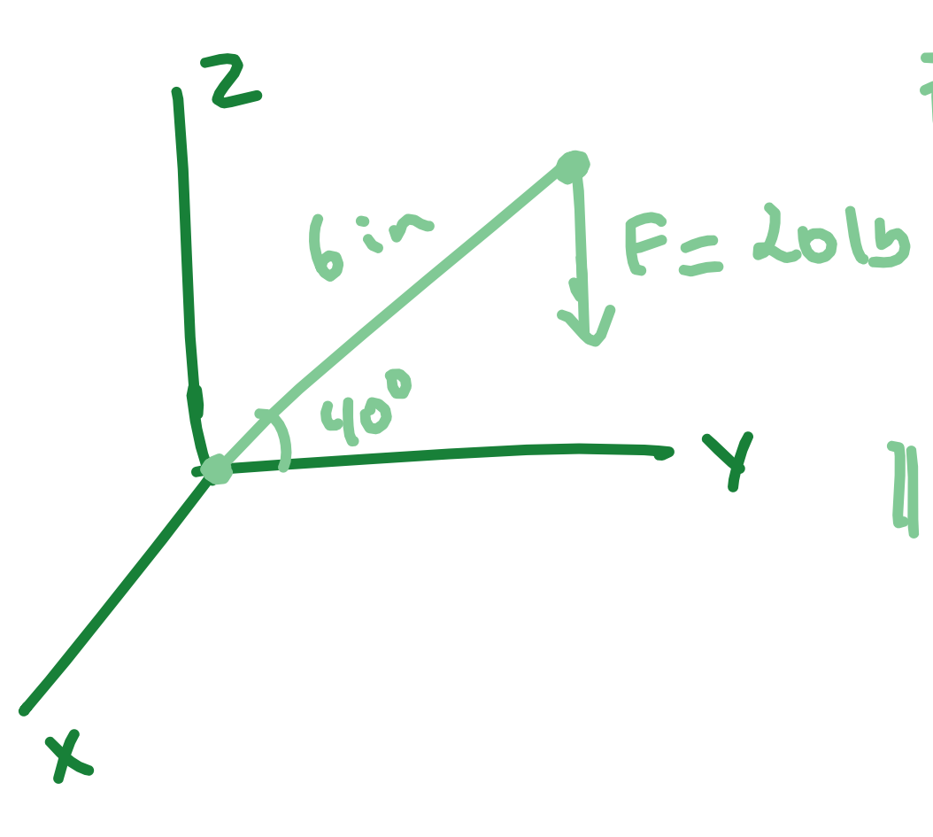
$\overrightarrow{PQ} = \langle 0, 1\cos 60°, 1\sin 60°\rangle$

$\vec{F} = \langle 0, 0, -50\rangle$

$\overrightarrow{PQ}\times\vec{F} = \langle -25, 0, 0\rangle$

$\|\overrightarrow{PQ}\times\vec{F}\| = 25$

Ex #29)



$\overrightarrow{PQ} = \langle 0, \frac{1}{2}\cos 40°, \frac{1}{2}\sin 40°\rangle$

$\vec{F} = \langle 0, 0, -20$

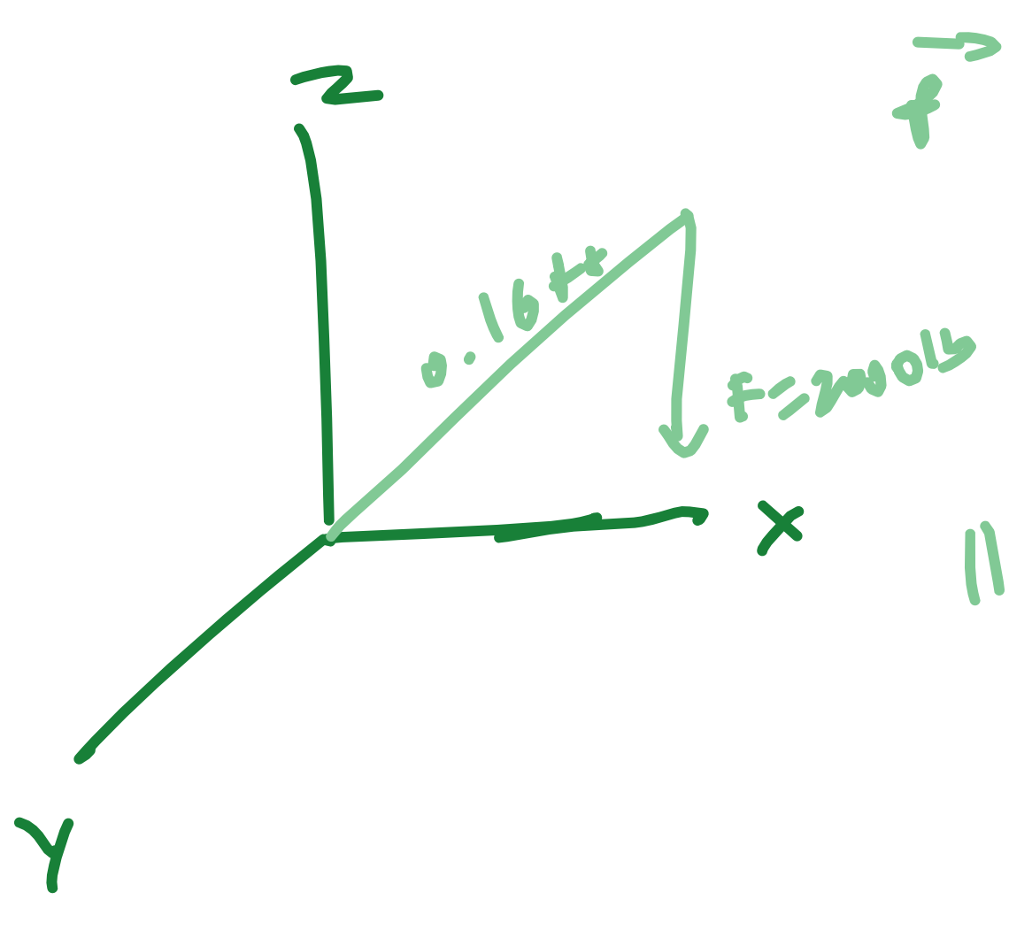
$\overrightarrow{PQ}\times\vec{F} = \langle -10\cos 40°, 0, 0\rangle$

$\|\overrightarrow{PQ}\times\vec{F}\| = 10\cos 40°$

Ex #30)

$\overrightarrow{PQ} = \langle 0, 0.16\cos 30°, 0.16\sin 30°\rangle$

$\vec{f} = \langle 0, 0, -2000\rangle$



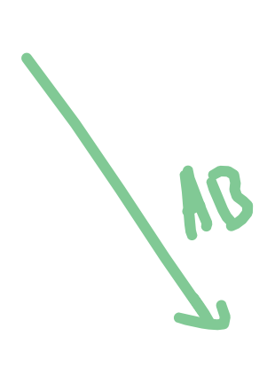
$\overrightarrow{PQ}\times\vec{F} = \langle -320\sin 30°, 0, 0\rangle$

$\|\overrightarrow{PQ}\times\vec{F}\| = 160$

Ex #31)

$\vec{F} = \langle -180\cos\theta, 0, -180\sin\theta\rangle$

$\times\ \overrightarrow{AB} = \langle -\frac{5}{4}, 0, 1\rangle$



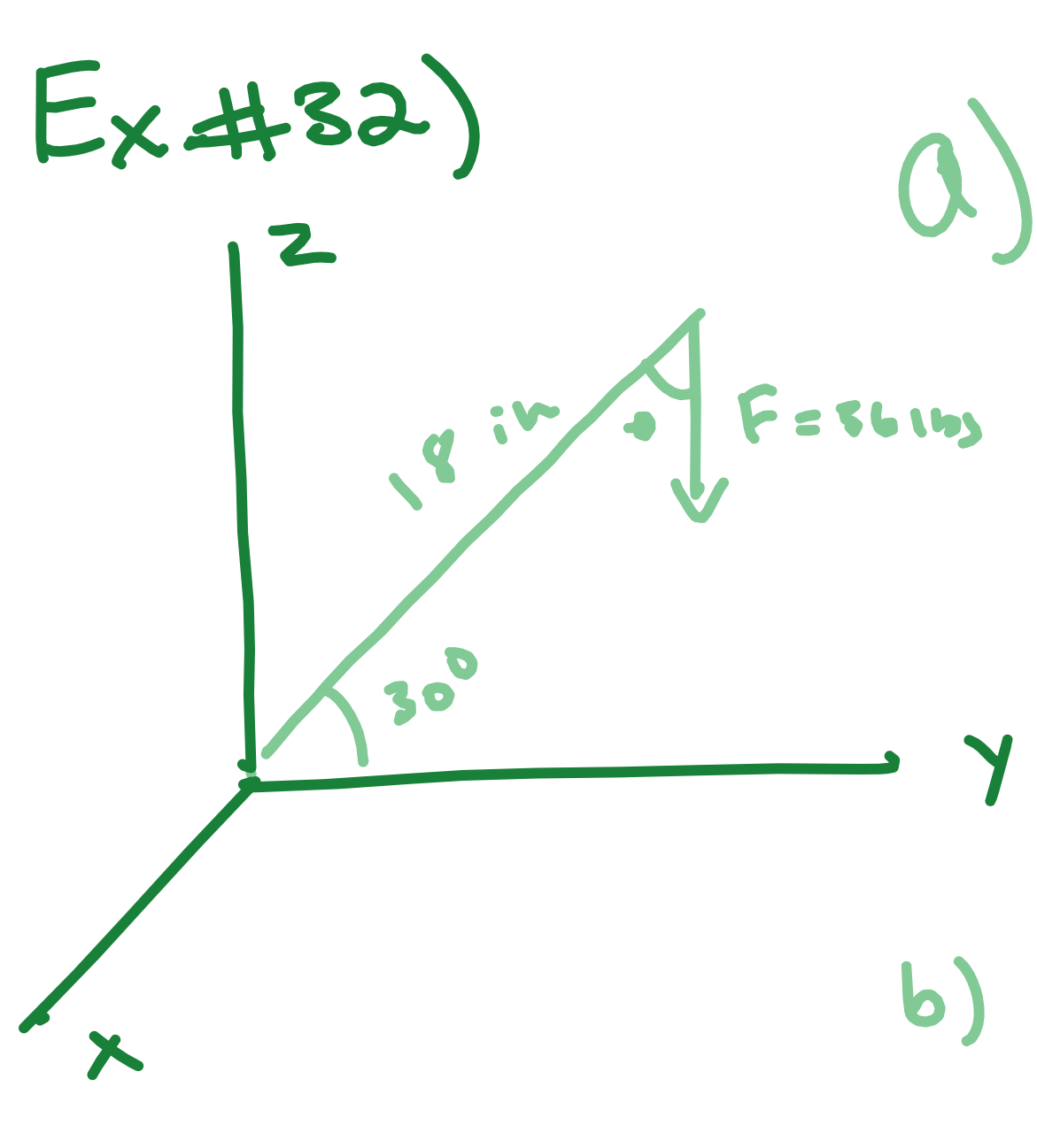
$\overrightarrow{AB}\times\vec{F} = \langle 0, 180\cos\theta, 180\cdot\frac{5}{4}\sin\theta\rangle$

$\|\overrightarrow{AB}\times\vec{F}\| = 180(\cos\theta + \frac{5}{4}\sin\theta)$

$180\left(\frac{\sqrt{3}}{2} + \frac{5}{8}\right)^2$

$\frac{48 + 40\sqrt{3} + 25}{64}$

Ex #32)



a) $\overrightarrow{OA} = \langle 0, \frac{3}{2}\cos 30°, \frac{3}{2}\sin 30°\rangle$

$F = \langle 0, 0, -56\rangle$

$\overrightarrow{OA}\times F = \langle 42\sqrt{3}, 0, 0\rangle$

$\|\overrightarrow{OA}\times F\| = 42\sqrt{3}$

b) $42\sqrt{2}$

c) $\theta = 0$ to maximize $\cos\theta$

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#29, 30, 31, 32 above as examples ↑

47) False, it needs to be 3D

48) False, should be opposite signs

49) False, $u = \langle 1, 1, 1\rangle$ $v = \langle 1, 2, 3\rangle$ $w = \langle 3, 2, 1\rangle$

50) True